

YANGYANG INTL, Korea, Republic of

WMO: 470920

Lat: 38.061N

Lon: 128.669E

Elev: 74

StdP: 100.45

Time Zone: 9.00 (E09)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-6.9	-24.8	0.4	-5.6	-23.0	0.5	-4.9	13.1	-2.5	12.0	-1.8	7.9	260	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	31.8	23.9	30.1	23.4	29.0	22.5	25.4	28.8	24.7	28.2	24.0	27.6	4.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.3	26.8	23.8	18.8	26.6	22.9	17.8	26.0	78.0	28.5	75.1	28.2	72.4	27.9	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.8	8.6	-11.4	34.2	2.5	1.1	-13.2	35.0	-14.6	35.7	-16.0	36.3	-17.8	37.2		
(5)			-13.3	26.0	2.2	0.9	-14.9	26.7	-16.3	27.2	-17.5	27.7	-19.1	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	0.3	1.8	6.3	11.9	17.0	20.0	23.9	24.5	19.9	15.4	9.1	2.3	(6)
(7)		DBStd	9.08	3.62	3.71	4.35	4.25	3.95	3.20	3.61	2.89	2.27	2.92	4.10	4.19	(7)
(8)		HDD10.0	996	301	229	133	26	1	0	0	0	0	2	64	241	(8)
(9)		HDD18.3	2565	559	462	373	198	75	17	1	0	8	98	276	498	(9)
(10)		CDD10.0	2004	0	1	18	83	216	301	431	448	298	169	39	1	(10)
(11)		CDD18.3	531	0	0	1	5	32	67	173	190	56	7	0	0	(11)
(12)		CDH23.3	3453	0	0	1	26	172	342	1363	1409	133	8	0	0	(12)
(13)		CDH26.7	953	0	0	0	4	24	69	461	381	14	1	0	0	(13)
(14)	Wind	WSAvg	3.8	5.0	4.4	4.3	4.2	3.6	2.6	2.7	2.7	3.4	4.3	5.1	(14)	
(15)	Precipitation	PrecAvg	1473	45	49	58	72	101	133	275	306	233	86	75	39	(15)
(16)		PrecMax	1994	90	144	142	156	330	316	670	685	590	268	238	198	(16)
(17)		PrecMin	1104	2	2	6	18	14	23	102	67	26	11	5	1	(17)
(18)		PrecStd	269	27	37	38	40	64	70	148	151	152	73	60	47	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	13.1	21.1	26.2	29.1	31.0	33.9	33.2	28.2	24.2	20.1	13.1	(19)
(20)			MCWB	4.4	6.5	12.1	14.1	16.4	19.6	24.6	24.8	20.4	15.3	12.9	7.4	(20)
(21)		2%	DB	8.1	10.2	18.0	23.0	27.0	28.2	32.0	31.2	26.1	22.1	18.0	11.1	(21)
(22)			MCWB	3.4	4.8	10.0	12.3	16.1	18.8	24.1	24.4	20.0	15.2	11.8	6.8	(22)
(23)		5%	DB	6.8	8.9	15.2	20.9	25.1	26.9	30.8	30.0	24.8	21.0	16.8	9.2	(23)
(24)			MCWB	2.2	3.8	8.0	11.4	15.0	18.9	24.0	24.1	19.8	15.3	11.5	4.9	(24)
(25)		10%	DB	5.2	7.1	13.0	18.8	23.8	25.1	29.1	28.9	23.8	20.0	15.1	8.0	(25)
(26)			MCWB	0.9	2.6	6.4	10.4	14.5	18.3	23.2	23.4	19.5	14.9	10.3	4.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.7	13.1	15.3	19.1	22.2	26.2	26.3	22.9	19.6	15.2	9.6	(27)
(28)			MCD B	8.4	11.1	19.9	23.4	26.0	26.8	29.8	29.5	25.3	20.6	17.3	11.6	(28)
(29)		2%	WB	4.1	5.5	10.6	13.9	17.5	21.1	25.2	25.7	21.7	17.7	13.5	7.0	(29)
(30)			MCD B	6.9	8.8	16.6	20.4	23.6	24.7	29.7	28.7	24.2	20.1	16.3	10.4	(30)
(31)		5%	WB	2.7	4.4	8.8	12.6	16.4	20.5	24.5	25.0	20.9	16.7	12.5	5.4	(31)
(32)			MCD B	5.5	7.6	13.7	18.2	22.1	23.8	29.1	28.0	23.4	19.4	15.6	8.4	(32)
(33)		10%	WB	1.6	3.3	7.3	11.4	15.6	19.7	23.8	24.4	20.3	15.8	11.1	4.2	(33)
(34)			MCD B	4.4	6.4	11.6	16.5	20.8	22.9	28.1	27.5	22.7	18.8	14.4	7.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.0	6.5	7.1	7.5	7.5	5.9	5.0	5.1	5.7	6.5	6.4	6.3	(35)
(36)			MCDBR	6.8	8.1	9.7	10.4	10.2	8.9	6.9	6.6	6.9	7.5	7.0	6.7	(36)
(37)			MCWBR	5.1	5.7	5.9	5.6	4.7	3.6	2.9	3.0	4.0	4.9	4.9	5.4	(37)
(38)		5% WB	MCDBR	6.0	7.3	9.2	9.4	8.8	6.5	6.3	5.6	5.4	6.1	6.0	6.2	(38)
(39)			MCWBR	5.0	5.6	6.2	5.8	4.8	3.3	3.1	3.3	3.7	4.7	4.8	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.409	0.471	0.562	0.608	0.614	0.662	0.628	0.555	0.485	0.480	0.445	0.401	(40)	
(41)		taud	1.995	1.870	1.722	1.677	1.713	1.677	1.813	1.999	2.119	1.996	2.014	2.043	(41)	
(42)		Ebn at Noon	723	725	698	698	706	671	691	733	763	710	679	698	(42)	
(43)		Edh at Noon	131	169	216	239	235	243	211	171	144	147	128	117	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.44	3.21	4.10	4.91	5.44	5.17	4.28	4.25	3.92	3.40	2.40	2.16	(44)	
(45)		RadStd	0.22	0.32	0.40	0.40	0.54	0.42	0.59	0.54	0.47	0.29	0.32	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	+55

Nomenclature: See separate page